

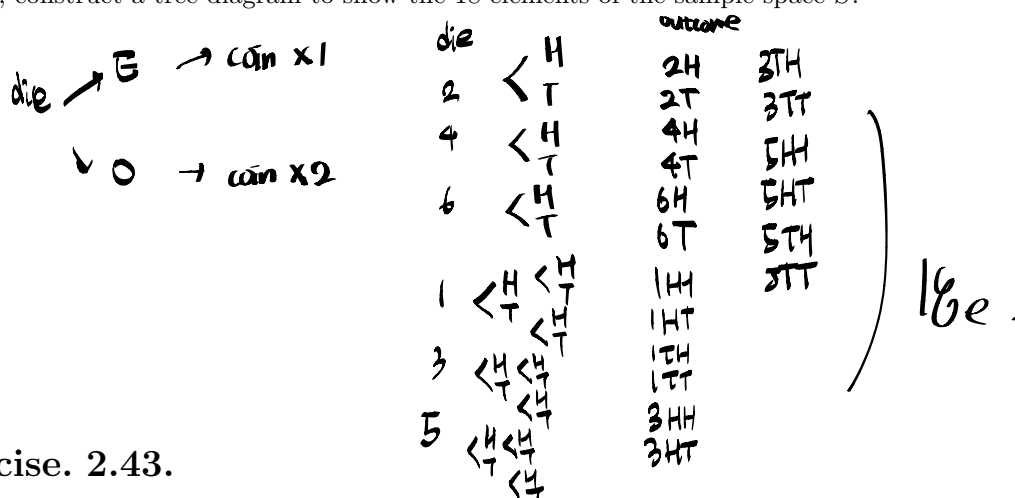
MATH230 HW1

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Exercise. 2.5.

An experiment consists of tossing a die and then flipping a coin once if the number on the die is even. If the number on the die is odd, the coin is flipped twice. Using the notation $4H$, for example, to denote the outcome that the die comes up 4 and then the coin comes up heads, and $3HT$ to denote the outcome that the die comes up 3 followed by a head and then a tail on the coin, construct a tree diagram to show the 18 elements of the sample space S .



Exercise. 2.43.

In how many ways can 6 different trees be planted in a circle?

A B C D E F.
 = B C D E F A
 C D E F A B
 D E F A B C
 E F A B C D
 F A B C D E

same... so

2) $\frac{6!}{6} = 5! = 5 \times 4 \times 3 \times 2 \times 1 = 120.$

Exercise. 2.45.

How many distinct permutations can be made from the letters of the word INFINITY?

3I 2N 1F 1T 1Y = 8.

1 8

so $\binom{8}{3,2,1,1,1} = \frac{8!}{3!2!} = \frac{8 \times 7 \times 6 \times 5 \times 4}{2} = 3360.$

Exercise. 2.58.

A pair of fair dice is tossed. Find the probability of getting

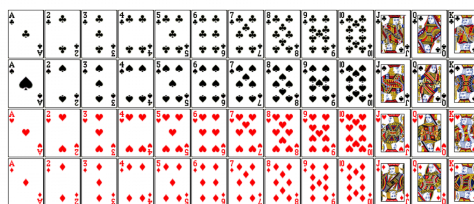
(a) a total of 9.

(b) at most a total of 3.

$$P(A) = \frac{n(A)}{n(S)} = \frac{4}{36} = \frac{1}{9}.$$

$$\sum = 2 \text{ or } \sum = 3 \Rightarrow \frac{3}{36} = \frac{1}{12}.$$

⑦ equally likely.



Exercise. 2.59.

In a poker hand consisting of 6 cards, find the probability of holding

(a) 3 aces;

(b) 4 hearts and 2 clubs.

$$\frac{4C_3 \times 48C_3}{52C_6} \approx 0.0034 = 0.34\%$$

$$\hookrightarrow \frac{13C_4 \times 13C_2}{52C_6} \approx 0.27\%.$$

Exercise. 2.74.

A class in advanced physics is composed of 10 juniors, 30 seniors, and 10 graduate students. The final grades show that 5 of the juniors, 10 of the seniors, and 5 of the graduate students received an A for the course. If a student is chosen at random from the class and is found to have earned an A, what is the probability that he or she is a senior?

$$P(B_2 | A) = \frac{P(A \cap B_2)}{P(A)} = \frac{\frac{1}{5}}{\frac{2}{5}} = \frac{1}{2}.$$

Exercise. 2.80.

The probability of an automobile being filled with gasoline also needing an oil change is 0.25; the probability that it needs a new oil filter is 0.40; and the probability that both the oil and the filter need changing is 0.2.

(a) If the oil has to be changed, what is the probability that a new oil filter is needed?

(b) If a new oil filter is needed, what is the probability that the oil has to be changed?

$$P(A|B) = \frac{0.2}{0.4} = \frac{1}{2} \quad \text{or} \quad \frac{0.2}{0.4} = \frac{1}{2} = 0.5$$

Exercise. 2.96.

Police plan to enforce speed limits by using radar traps at four different locations within the city limits. The radar traps at each of the locations L_1, L_2, L_3, L_4 will be operated 40%, 30%, 20%, and 30% of the time. If a person who is speeding on her way to work has probabilities of 0.2, 0.1, 0.5, and 0.2, respectively, of passing through these locations, what is the probability that she will receive a speeding ticket?

$$P(B) = \sum_{i=1}^4 P(B|A_i) P(A_i) = 0.2 \times 0.4 + 0.1 \times 0.3 + 0.5 \times 0.2 + 0.2 \times 0.3 = 0.08 + 0.03 + 0.10 + 0.06 = 0.27$$

Exercise. 2.98.

If the person in Exercise 2.96 received a speeding ticket on her way to work, what is the probability that she passed through the radar trap located at L_2 ?

$$P(A_2|B) = \frac{P(A_2 \cap B)}{P(B)} = \frac{P(B|A_2) P(A_2)}{P(B)} = \frac{0.3 \times 0.1}{0.27} = \frac{0.03}{0.27} = \frac{1}{9}$$